# WASP-43b

**R-1545**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_220109` · created 2026-08-01T19:29:31+00:00

**BENCHMARK: NON-DETECTION**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2459588.9213 +/- 0.0021 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.137 +/- 0.056 |
| Transit depth (Rp/Rs)^2 | 1.87 +/- 1.54 [%] |
| Orbital Inclination (inc) | 85.44 +/- 2.94 |
| Airmass coefficient 1 (a1) | 0.99 +/- 0.64 |
| Airmass coefficient 2 (a2) | -0.02 +/- 1.41 |
| Scatter in the residuals of the lightcurve fit is | 62.64 % |
| Best Comparison Star | None |
| Optimal Aperture | 2.62 |
| Optimal Annulus | 13.98 |
| Transit Duration (day) | 0.0554 +/- 0.0073 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.137 ± 0.056 vs 0.1594 (deviation 0.4σ) |
| Duration fit vs published | 0.0554 d vs 0.0512 d (deviation 0.57σ) |
| Mid-time O–C | 5.1 min |
| Depth SNR | 2.4 |
| Residual scatter | 62.64 % |

## Light curve

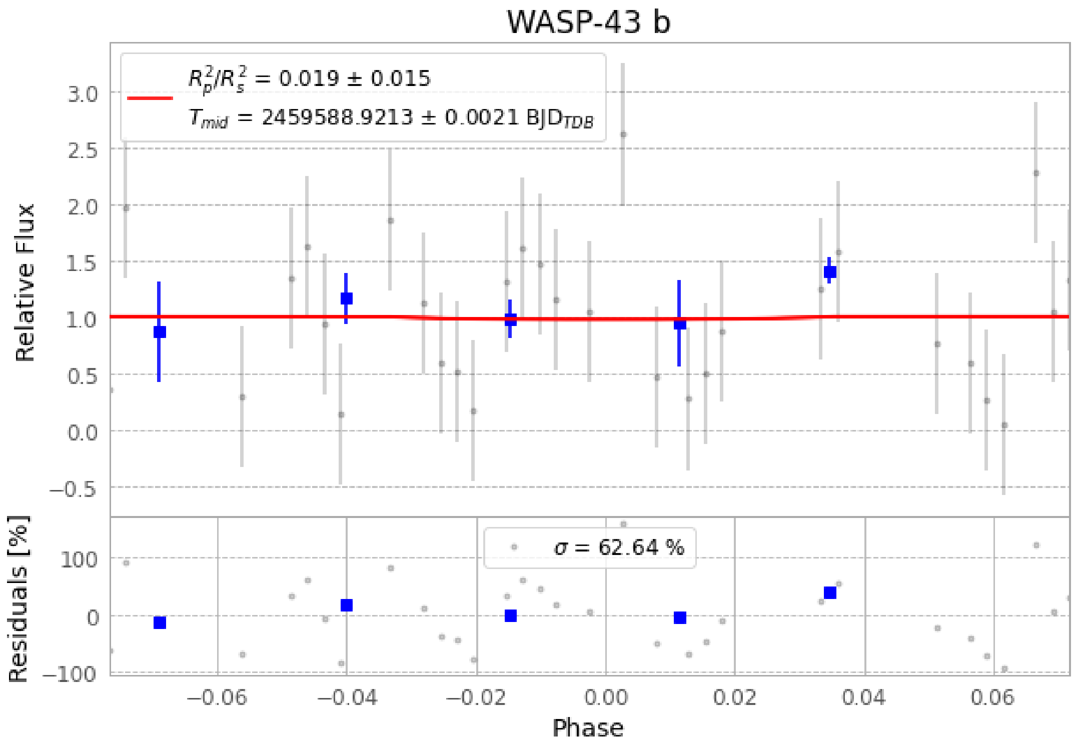


Figure 1 — FinalLightCurve_WASP-43 b_2022-01-09.png (SHA-256 98881120c8e2a641…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 181], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 02aae1a59c2aa9d1…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

## Data provenance

61 FITS frames (40 MB), `WASP-43220109083012.FITS` → `WASP-43220109113012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220109.log (edc9709a1477…), FinalLightCurve_WASP-43 b_2022-01-09.png (98881120c8e2…), FinalLightCurve_WASP-43 b_2022-01-09.csv (09ad3514cd4c…)

## Compute resources

Wall time 195.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)